Drop Shadow FX

Alpha-based shadows, inset shading, and shadow-only outputs

Drop Shadow FX takes the alpha silhouette of a connected Source image, blurs it, offsets it by an Angle and Distance, and colors the result. The Source may remain visible above the shadow, or the effect may output only the shadow.

THREE MAIN USES

Outer shadow

An offset shade behind cutout art.

Inner shadow

Recessed shading within the original.

Shadow-only

A separate compositing matte.

QUICK START

1. Open FX Schematic. Add Light > Drop Shadow and connect the artwork to the Source port.

2. Start with default black Color; Blur X/Y = 2, Quality = 1, Strength = 1, Angle = 45, Distance =
2. Inner, Knockout and Hide Object begin off.

3. Increase Distance for a detached shadow. Increase Blur X/Y for softness. Adjust Color alpha and Strength for opacity.

4. Enable Inner for inset shade; Hide Object to output only shade; Knockout for an outer shade excluding the original silhouette.

WORKING WITH THE FX NODE

The effect reads the Source alpha, so transparent artwork gives a distinct outer shadow. Composite a shadow-only branch beneath the original artwork for independent editing. A fully opaque source covering the entire frame leaves no visible exterior edge.

Controls and behavior

FX ID: STD_dropShadowFx | Category: Light | Source: one image input

CONTROL	DEFAULT	FUNCTION
Color	Black	Colored tint of the alpha shadow, including color opacity.
Blur X / Y	2 / 2	Independent horizontal and vertical blur; FX lengths.
Quality	1 (0-15)	Blur passes. A value of zero skips blurring.
Strength	1 (0-255)	Scales shade intensity; resulting opacity is clamped.
Angle	45 degrees	Direction of offset: 0 right, 90 down in output space.
Distance	2	Offset of shadow. Zero aligns it with Source.
Inner	Off	Reverse blurred silhouette, clipped to source alpha.
Knockout	Off	Outer: shadow outside the source alpha; hide object.
Hide Object	Off	Suppress original art, leaving a shadow-only output.

COMPOSITION MODES

- Outer, switches off: original artwork sits in front of the offset shade.
- Outer + Hide Object: shade only, including shade beneath the source silhouette.
- Outer + Knockout: shade only outside the Source silhouette; artwork omitted.
- Inner: shade restricted to the Source alpha, normally overlaid on source art.
- Inner + Hide Object or Knockout: inset shading only, original omitted.

Animating: Color, Blur X/Y, Strength, Angle and Distance are sampled at the current frame. Quality and mode switches are read without frame sampling. Offsets are rounded to rendered pixels, so small movements can step by one pixel.

Comparison with other FX

Similar-looking results use different source data and compositing behaviors

EFFECT	KEY DIFFERENCE	USE WHEN
DROP SHADOW	One color, offset alpha matte outside or inside.	Cutouts, titles, character separation, inset shade.
BEVEL	Two-sided edge lighting: Highlight + Shadow. Inner, Outer, Full.	Raised or engraved type; surface edge relief.
BODY HIGHLIGHT	Tinted interior overlay, point X/Y, blur, invert, blend modes.	Interior contour highlight or body shading.
CAST SHADOW	Projected, distortable shadow with top/bottom blur/opacity.	Ground or wall shadow with perspective.
BACKLIT	Separate Light input affects Source.	Image-driven lighting from a second layer.
GLOW / EMBOSS	Light halo / surface relief rather than displaced silhouette.	Luminous bloom / texture-depth treatment.

DROP SHADOW VS. BEVEL

Both new effects use shared raster and Flash-style box-blur utilities. Drop Shadow produces a single colored matte moved by Angle and Distance. Bevel samples opposite offset sides of the alpha silhouette to form simultaneous highlight and dark edges.

Bevel supports Inner, Outer and Full and distinct Highlight and Shadow colors; Drop Shadow instead offers Inner and the shadow-only Hide Object / Knockout controls. A bevel makes a shape appear raised or engraved, not simply separated from its background.

Quick choice: need a detached outline shadow? Drop Shadow. Need dual light/dark relief? Bevel. Need tint inside a drawing with blend controls? Body Highlight. Need a skewed projection onto a floor?

Practical recipes

Suggested starting points, not built-in presets

01 CARTOON SEPARATION

Outer, all switches off. Choose a cool dark Color, a modest Distance, Blur X/Y around 3-6 and Quality around 1-2. Keep intensity subtle.

02 INSET GRAPHIC

Inner on. Use a darker relative of the fill, short Distance and small blur. Ideal for lettering, buttons and recesses.

03 SEPARATE SHADOW PASS

Inner off; Hide Object on. Route output into a branch below the original drawing. Keep Knockout off if the full shadow matte is needed.

04 CRISP CUTOUT

Set Blur X/Y to zero or Quality to zero for a sharp graphic silhouette. Try multiple passes through distinct Drop Shadow nodes for layered offsets.

NOTES AND TROUBLESHOOTING

- No visible outer shade? Verify a Source is connected, alpha has transparent edges, and Color opacity / Strength are nonzero.

- Jagged or stepped movement? Increase blur as appropriate; angle/distance shifts intentionally round to rendered pixels.

- Different preview vs final? Blur and Distance scale with render transform. Confirm at the final output size.

- Knockout omits the original artwork. With an outer shadow it also removes shade where Source alpha is opaque.

SOURCE AND ATTRIBUTION

From OT-Dev PR #273, extracted from Flash/Adobe Animate import PR #267 by @doangelshavewings; original contributor attribution is preserved. Shared header shadowbevelutils.h also serves Bevel FX PR #274. Rendering supports 8-, 16- and float-raster paths. Interactive regression testing should cover scale, alpha, frame animation and scene save/reload.

github.com/OpenAnimationLibrary/opentoonz-dev/pull/273